

# EarthBridge: A Solution for the 4th Multi-modal Aerial View Image Challenge – Translation Track

**Team:** EarthBridge    **CodaBench:** bilisakura    **Code:** [github.com/Bili-Sakura/EarthBridge-Preview](https://github.com/Bili-Sakura/EarthBridge-Preview)

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## Abstract

We present **EarthBridge**, our solution for the 4th MAVIC-T challenge. We apply Diffusion Bridge Implicit Models (DBIM) [6] to three translation tasks (SAR→EO, SAR→RGB, RGB→IR) and Contrastive Unpaired Translation (CUT) for SAR→IR. Our solution achieves the second position overall. We report composite scores of 0.269, 0.576, 0.464, and 0.200 on SAR→EO, SAR→RGB, SAR→IR, and RGB→IR respectively, with an average of 0.38.

## 1. Method

### 1.1. Diffusion Bridge Implicit Models (DBIM)

**Stochastic Bridges.** Given a base diffusion process  $dx_t = f(x_t, t)dt + g(t)dw_t$ , the bridge conditioned to reach target  $x_0 = \mathbf{y}$  starting from source  $x_T = \mathbf{x}$  is:

$$dx_t = f(x_t, t)dt + g(t)dw_t + g(t)^2 \nabla_{x_t} \log p(x_0 = \mathbf{y} | x_t) dt. \quad (1)$$

Under a variance-preserving (VP) schedule, intermediate states are:

$$z_t = a_t \mathbf{x} + b_t \mathbf{y} + \sigma_t \epsilon, \quad \epsilon \sim \mathcal{N}(0, I), \quad (2)$$

where  $\mathbf{x}$  is the source image,  $\mathbf{y}$  is the target image, and  $a_t, b_t, \sigma_t$  are schedule coefficients [7].

### Non-Markovian Generalization and Booting Noise.

We extend the Markovian bridge to a family of non-Markovian processes parametrized by variance  $\rho \in \mathbb{R}^{N-1}$ . The reverse update rule is:

$$z_{t_n} = a_{t_n} \mathbf{x} + b_{t_n} \hat{z}_0 + \sqrt{c_{t_n}^2 - \rho_n^2} \hat{\epsilon} + \rho_n \epsilon, \quad (3)$$

where  $\hat{z}_0 = D_\theta(z_{t_{N+1}}, t_{N+1}, \mathbf{x})$  is the UNet denoiser prediction. Setting  $\rho_n = 0$  gives the deterministic DBIM [6], analogous to an Euler discretization of a probability-flow ODE over the bridge. Since  $c_{t_N} = 0$  leads to a singularity at initialization, we force the first step to be stochastic by setting  $\rho_{N-1} = c_{t_{N-1}}$ , injecting Gaussian booting noise that encodes the one-to-many mapping ambiguity.

**Training Objective.** The model is trained with Karras-weighted bridge loss:

$$\mathcal{L} = \mathbb{E}_{(\mathbf{x}, \mathbf{y}), t, \epsilon} [w(t) \cdot \|D_\theta(z_t, t, \mathbf{x}) - \mathbf{y}\|_2^2], \quad (4)$$

where  $w(t)$  follows Karras bridge weighting [2] and  $t$  is sampled from a Karras inverse- $\rho$  distribution.

**Architecture.** Our denoiser  $D_\theta$  is a UNet [5] that receives the channel-wise concatenation  $[z_t, \mathbf{x}]$  as input. Timestep information is injected via sinusoidal embeddings and Adaptive Group Normalization (AdaGN, ADM-style [1]). The architecture overview is shown in Fig. 1.

For tasks where source and target modalities have different channel counts (e.g., RGB→IR or SAR→RGB), we operate the bridge in a shared channel space (typically 3 for RGB-related tasks). Single-band modalities are expanded by channel repetition before entering the UNet, ensuring the concatenation  $[z_t, \mathbf{x}]$  remains well-defined across all tasks.

### 1.2. Contrastive Unpaired Translation (CUT)

For SAR→IR we use CUT [4]: patch-based contrastive learning preserves structure without pixel-aligned pairs. Config: Tab. 2.

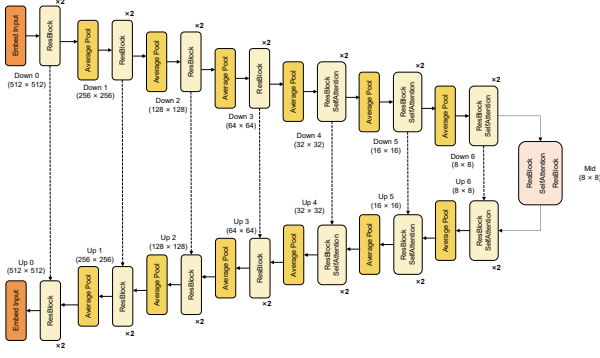


Figure 1. UNet architecture of our DBIM for 1024px tasks (SAR→RGB, RGB→IR). We train at 512×512 and infer at full resolution. Source  $x$  is channel-concatenated with noisy sample  $z_t$  and processed by this denoiser. Bridge scalings  $a_t, b_t, c_t$  construct  $z_t$  during training; deterministic DBIM sampling generates  $\hat{z}_0$  during inference.

## 2. Experiments

### 2.1. Dataset

SAR→EO uses 256×256 imagery with 89,411 pairs. The other three tasks use 1024×1024: SAR→RGB (10,576 orig., 75,703 aug.), SAR→IR (10,576 orig., 74,317 aug.), RGB→IR (2,272 orig., 7,388 aug.). For SAR→IR and SAR→RGB, we spatially align SAR scenes with optical mosaics before resampling. For 1024×1024 tasks, we augment training data with sliding-window crops (stride 512).

### 2.2. Training Details

All DBIM models are trained with the Prodigy optimizer [3]. For 1024×1024 tasks, we train at 512×512 and infer at full resolution. The bridge follows a VP log-SNR schedule with  $\beta_d = 2.0$ ,  $\beta_{\min} = 0.1$ . We use DBIM for SAR→EO, SAR→RGB, and RGB→IR; CUT for SAR→IR only. CUT is trained with Prodigy, batch size 2 per GPU across 8 GPUs, for 11 epochs, with NCE temperature 0.1 and  $\lambda_{\text{GAN}} = 0.5$ . Configurations are given in Tabs. 1 and 2.

Table 1. Architecture and training settings for DBIM models used in the SAR→EO, SAR→RGB, and RGB→IR tasks.

| Configurations        | SAR→EO           | SAR→RGB          | RGB→IR |
|-----------------------|------------------|------------------|--------|
| Base channels         | 64               | 96               | 128    |
| Channel multipliers   | (1,2,3,4)        | (1,1,2,2,4,4)    |        |
| Attention resolutions | N/A              | 32, 16, 8        |        |
| Training resolution   | 256 <sup>2</sup> | 512 <sup>2</sup> |        |
| Epochs                | 55               | 24               | 86     |

Table 2. Network architecture and optimization details for CUT model used in the SAR→IR task.

| Configurations                  | SAR→IR           |
|---------------------------------|------------------|
| Number of generator filters     | 256              |
| Number of discriminator filters | 256              |
| Generator architecture          | ResNet-9block    |
| Discriminator architecture      | 4-layer PatchGAN |
| Training resolution             | 512 <sup>2</sup> |
| Training epochs                 | 11               |

### 2.3. Hardware and Software

**Hardware.** A single node cluster with 8 NVIDIA H200 GPUs (140GB); x86\_64 multi-core CPU (AMD EPYC / Intel Xeon); 512 GB+ RAM.

**Software.** Python 3.10+; PyTorch 2.8; HuggingFace Diffusers 0.36; Transformers; Accelerate; Rasterio. Code: [github.com/Bili-Sakura/EarthBridge-Preview](https://github.com/Bili-Sakura/EarthBridge-Preview).

### 2.4. Quantitative Results

Tab. 3 reports the performance of our **EarthBridge** final submission. NFE denotes number of function evaluations.

Table 3. Results on the MAVIC-T evaluation set. ↓ lower is better.

| Track   | Method | FID↓  | LPIPS↓ | L1↓   | Score↓ | NFE  | Time   |
|---------|--------|-------|--------|-------|--------|------|--------|
| SAR→EO  | DBIM   | 0.222 | 0.504  | 0.080 | 0.269  | 500  | 0.42 s |
| SAR→RGB | DBIM   | 0.878 | 0.636  | 0.214 | 0.576  | 1000 | 160 s  |
| SAR→IR  | CUT    | 0.645 | 0.603  | 0.145 | 0.464  | 1    | 0.47 s |
| RGB→IR  | DBIM   | 0.357 | 0.151  | 0.090 | 0.200  | 5    | 1.5 s  |

**Key Observations.** (1) SAR→EO and RGB→IR are extremely easy to model; we achieve high performance on both test and validation sets, which aligns with peer rankings in previous years. (2) SAR→RGB and SAR→IR are quite difficult, also consistent with prior competition results. Given the difficulty of SAR→IR, we switched from DBIM to the GAN-based CUT. This yielded better results: diffusion bridges for this hard task tend to produce over-smooth, blurry outputs with low training loss that nonetheless deviate from the actual target appearance; CUT’s discriminator addresses this. (3) Although inference requires full 1024×1024 resolution, we find that training our UNet at 512×512 effectively extends to 1024×1024 inference without modification. (4) Our solution is computationally heavy: we employ models over 100M parameters and train on a node cluster with 8 NVIDIA H200 GPUs for hours.

## References

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